NDRA-PII

Neuro-Semantic Multi-Agent PII Intelligence System

"Identity risk emerges from meaning, not patterns"

Executive Overview

NDRA-PII represents a paradigm shift in how organizations approach Personally Identifiable Information (PII) detection and governance. Unlike traditional pattern-matching systems, NDRA-PII employs advanced neuro-semantic reasoning to understand context, infer implicit identities, and provide explainable, policy-driven protection for sensitive data across diverse document types and formats.

> **Core Innovation:** NDRA-PII treats identity as a semantic intelligence problem, combining Microsoft Presidio's robust detection engine with multi-agent AI orchestration, distributed vector stores, and neuro-symbolic reasoning to achieve unprecedented accuracy and explainability in PII governance.

System Architecture

NDRA-PII is built on a microkernel-based, multi-agent architecture where specialized agents collaborate through message-passing to achieve comprehensive PII intelligence. The system operates across eight distinct architectural layers:

```
Users (Web UI / CLI / Email / API) ↓ Auth Layer (SSO / OAuth2 / MFA) ↓ API Gateway (Rate Limiting / WAF) ↓
┌──────────────────────────────────────────┐ │ Document Ingestion Layer │ │ - Upload Worker │ │ - Email
Harvester │ │ - Streaming Connectors │ └──────────────────────────────────────────┘ ↓
┌──────────────────────────────────────────┐ │ Document Intelligence Layer │ │ - OCR (Tesseract/PaddleOCR)
│ │ - Layout Analysis (LayoutLM) │ │ - Semantic Chunker │ └──────────────────────────────────────────┘ ↓
┌──────────────────────────────────────────┐ │ PII Detection Layer │ │ - Presidio AnalyzerEngine │ │ -
Custom Recognizers │ │ - NER Models │ │ - LLM Evidence Extraction │
└──────────────────────────────────────────┘ ↓ ┌──────────────────────────────────────────┐ │ Multi-
Agent Orchestration │ │ (Kafka Message Bus) │ └──────────────────────────────────────────┘ ↓
┌──────────────────────────────────────────┐ │ Decision Engine (NSRL) │ │ - Risk Scoring │ │ - Policy
Enforcement │ │ - Explainability │ └──────────────────────────────────────────┘ ↓ Audit Ledger / UI
Dashboard / SIEM
```

Core Capabilities

🔍 Multi-Modal Detection

Processes PDFs, DOCX, emails, images, scanned documents, and streaming data with layout-aware OCR and semantic understanding.

🧠 Contextual Intelligence

Employs transformer-based models to detect implicit PII, inferred identities, and compositional risk beyond simple pattern matching.

⚖️ Neuro-Symbolic Reasoning

Combines deterministic rules with LLM-based inference for transparent, auditable decision-making that respects regulatory complexity.

🎯 Dynamic Risk Scoring

Calculates real-time risk scores (0-100) based on PII density, exposure context, jurisdictional requirements, and historical patterns.

🔐 Policy Enforcement

Supports GDPR, HIPAA, PCI-DSS, DPDP (India), and custom organizational policies with automated redaction and escalation workflows.

📊 Full Explainability

Provides line-level highlights, decision timelines, clause traceability, and cryptographically signed audit trails for compliance.

Multi-Agent System

NDRA-PII employs specialized agents, each with strictly limited authority and clear responsibilities. All inter-agent communication occurs through secure message queues with comprehensive logging.

Preprocessing Agent

Performs OCR, layout extraction, and semantic-aware chunking with coordinate mapping for provenance.

Embedding Agent

Dynamically selects optimal embedding models (MiniLM/LegalBERT/BioBERT) based on document domain and vectorizes content for retrieval.

Retrieval Agent

Executes hybrid BM25 + dense vector search with reranking to identify contextually relevant document sections.

PII Extractor Agent

Integrates Presidio AnalyzerEngine with custom recognizers for jurisdiction-specific IDs (Aadhaar, PAN, passport, etc.).

PII Fusion Agent

Merges detections from Presidio, NER models, and LLM inference while deduplicating overlapping spans.

NSRL Reasoner Agent

Applies compiled policy rules using a deterministic engine with LLM fallback for ambiguous cases.

Risk Analyzer Agent

Computes multi-dimensional risk scores using heuristics and ML ensemble models.

Redaction Agent

Leverages Presidio AnonymizerEngine to mask, hash, or tokenize sensitive information per policy.

Explainability Agent

Constructs decision timelines, highlights source evidence, and generates human-readable justifications.

Test Generator Agent

Automatically synthesizes positive/negative test cases from policy rules for continuous validation.

Microsoft Presidio Integration

NDRA-PII deeply integrates Microsoft Presidio as its foundational PII detection layer, dramatically reducing development complexity while ensuring production-grade accuracy:

Built-in Recognizers

Aadhaar | PAN | Passport | Credit Cards | Email Addresses | Phone Numbers | IP Addresses

Bank Accounts | Names | Addresses | SSN (US) | Medical IDs

Anonymization Strategies

Presidio's AnonymizerEngine supports multiple strategies: redact (complete removal), mask (XXX-XX-1234), hash (SHA-256/HMAC), encrypt (reversible tokenization), and replace (synthetic data substitution). NDRA-PII selects strategies dynamically based on document type, sensitivity level, and organizational policy.

Neuro-Symbolic Rule Layer (NSRL)

NSRL combines the precision of symbolic logic with the flexibility of neural inference. Rules are authored in a human-readable DSL and compiled to executable forms:

```
rule_id: R_HIGH_RISK_AADHAAR priority: 100 description: "Aadhaar in contract triggers escalation" when: -
pii.subtype == "aadhaar" - doc.type == "contract" - doc.metadata.policy_age_months < 6 then: - severity:
high - action: redact_and_escalate - notification: security_ops - retention_override: 7_years
```

The NSRL engine evaluates deterministic rules first, falling back to LLM-based interpretation for ambiguous cases. All rule firings are logged with cryptographic signatures for audit purposes.

Performance Metrics & KPIs

Metric	Target	Notes
PII Detection Precision	≥ 95%	For regulated identifiers (Aadhaar, PAN, SSN)
PII Detection Recall	≥ 92%	Across all PII categories
False Positive Rate	≤ 5%	Adjustable per tenant sensitivity
End-to-End Latency	< 8 seconds	For 10-page document (MVP target)
Rule Coverage	≥ 80%	Deterministic rules vs LLM fallback
Redaction Accuracy	≥ 99%	Complete PII removal without data corruption
Audit Completeness	100%	All decisions logged with provenance

Technology Stack

Backend Infrastructure

FastAPI | Python 3.10+ | Kubernetes | Kafka | PostgreSQL | Redis

AI & ML Components

Microsoft Presidio | HuggingFace Transformers | Sentence-BERT | LayoutLMv3 | Gemini/Llama

Vector & Storage

Milvus | Chroma | Neo4j | S3/GCS

Frontend & UI

Next.js | React | TypeScript | TailwindCSS | PDF.js

Security & Compliance

Security Principles

NDRA-PII implements defense-in-depth with zero-trust architecture:

- Agent isolation with no shared memory
- Mutual TLS for all inter-service communication
- Envelope encryption with per-tenant keys (KMS)
- PII redaction in all logs and metrics
- Role-based access control (RBAC) with step-up authentication
- Cryptographically signed audit trails (WORM ledger)

Regulatory Support

NDRA-PII provides pre-built policy templates and enforcement mechanisms for:

- **GDPR** (EU): Right to erasure, data portability, purpose limitation
- **HIPAA** (US Healthcare): PHI protection, minimum necessary standard
- **PCI-DSS**: Cardholder data security requirements
- **DPDP Act** (India): Data principal rights, consent management
- **CCPA/CPRA** (California): Consumer privacy rights

Data Subject Access Requests (DSAR)

Built-in DSAR workflows enable automated export of all data associated with an individual, including audit trails, to comply with right-to-access requirements.

Deployment Architecture

NDRA-PII supports flexible deployment models tailored to organizational requirements:

Cloud-Native SaaS

Multi-tenant deployment on AWS/GCP/Azure with per-tenant data isolation, global CDN distribution, and elastic autoscaling. Ideal for organizations seeking rapid deployment without infrastructure management.

On-Premises Enterprise

Self-contained Kubernetes distribution with air-gapped operation, hardware security module (HSM) integration, and full data sovereignty. Suitable for highly regulated industries (defense, finance, healthcare).

Hybrid & Edge

Distributed architecture with edge processing nodes for latency-sensitive applications and central control plane for policy management and analytics.

Development Timeline

Hackathon MVP (48-72 hours)

Basic document ingestion, Presidio integration, simple UI with PII highlights, JSON decision output. Demonstrates core detection and redaction capabilities.

Production Demo (2-4 weeks)

Multi-agent orchestration, dynamic embeddings, risk scoring, NSRL basic rules, explainability timeline, admin dashboard. Suitable for investor/enterprise demonstrations.

Enterprise Platform (4-6 months)

Full feature set including multi-tenancy, distributed vector store, comprehensive policy library, ML-based risk modeling, continuous learning from feedback, observability suite, penetration testing, and regulatory audit compliance.

Use Cases & Applications

🏦 Financial Services

Detect PII in loan applications, KYC documents, transaction records, and customer communications to ensure PCI-DSS and banking regulation compliance.

🏥 Healthcare

Identify PHI in medical records, insurance claims, discharge summaries, and research documents for HIPAA compliance and clinical trial safety.

🔐 Cybersecurity

Monitor data exfiltration attempts, scan logs for credential leaks, and detect inadvertent PII exposure in incident response workflows.

📧 Enterprise Email

Analyze email attachments and threads for sensitive information before external transmission, preventing accidental data breaches.

📜 Legal & Compliance

Redact confidential information from contracts, court filings, and regulatory submissions while maintaining document utility.

🏢 HR & Recruitment

Anonymize resumes and performance reviews for unbiased evaluations and protect employee personal information in workforce analytics.

Competitive Advantages

Why NDRA-PII?

Traditional PII systems rely on regex patterns and static rules, failing to understand context, infer implicit identities, or adapt to

evolving threats. NDRA-PII represents a fundamental advancement by treating identity as a semantic intelligence problem—combining the robustness of Presidio with cutting-edge neuro-symbolic AI to deliver explainable, policy-driven protection that scales from startups to Fortune 500 enterprises.

- **Context-Aware Detection:** Understands that "CEO of hospital in Kochi" is re-identification risk
- **Explainable Decisions:** Every determination traced to specific rules and evidence
- **Continuous Improvement:** Feedback loops enable learning from false positives/negatives
- **Regulatory Agility:** Adapt to new privacy laws through DSL rule updates, not code rewrites
- **Enterprise Ready:** Multi-tenancy, SSO, audit trails, and SLAs from day one

Future Roadmap

Near-Term (6-12 months)

- Cross-modal PII detection (audio transcripts, video OCR)
- Real-time streaming pipeline integration
- Advanced ML risk models with behavioral analysis
- Expanded recognizer library (50+ global ID formats)

Medium-Term (12-24 months)

- Federated learning for privacy-preserving model improvement
- Quantum-resistant cryptographic signing
- Automated policy synthesis from regulatory text
- Predictive compliance analytics (pre-breach risk identification)

Long-Term Vision

Transform NDRA-PII into a universal identity intelligence platform capable of protecting privacy across all data modalities, reasoning about complex multi-hop inferences, and serving as the trust layer for next-generation AI systems operating in regulated environments.

Getting Started

NDRA-PII is designed for rapid prototyping with tools like Google Antigravity IDE, which can auto-generate agent scaffolds, API endpoints, and UI components from the architectural specification. For production deployment, comprehensive documentation, Helm charts, and reference implementations are available.

Contact & Contributions:

NDRA-PII is an open architecture designed for academic research, enterprise adoption, and collaborative development. For implementation guidance, custom deployment support, or research partnerships, please reach out through the project repository or contact the core development team.